

Efficient Autoencoding with Quadtree-based Visual Transformer Encoder

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Abstract. Visual Transformer (ViT) architectures are now widely used in image processing. However, standard ViT autoencoders rely on uniform patchification, assigning the same computational budget to highly textured foregrounds and homogeneous backgrounds. This redundancy leads to unnecessary computation. A compute-efficient autoencoder framework is proposed that pairs an Adaptive Patch Tokenization (APT) encoder with a ViT decoder operating on a restored dense lattice. An entropy-based quadtree partitions the image into fine patches in complex regions and coarse patches in uniform regions, reducing the encoder sequence length. During reconstruction, the decoder inserts learned mask tokens to restore the dynamic sequence to the original spatial grid, allowing standard ViT transformer blocks to operate without decoder redesign. Quantitative results show that the proposed method reaches 36.21 GFLOPS, a 17.5% reduction relative to the ViT-B baseline, while maintaining competitive PSNR and SSIM.

Keywords. Autoencoder, Quadtree, ViT

1. Introduction

Computational efficiency has long been an important goal in computer vision. Earlier approaches worked under strict hardware constraints and therefore emphasized selective processing over brute-force treatment of every pixel. Techniques such as contour tracing, morphological operations, and hierarchical spatial partitioning [8] reduced computation by focusing resources on regions of interest. Modern deep learning models, particularly Visual Transformer (ViT) architectures [4], instead rely on parallel but expensive computation over dense, uniform grids. This design choice has supported strong performance, but often at the cost of efficiency [10]. As deployment in resource-constrained environments becomes more important, revisiting such efficiency-oriented ideas is well motivated.

A fundamental inefficiency in standard ViT architectures lies in their uniform patchification paradigm. An input image is divided into a rigid grid of equal-sized, non-overlapping patches, meaning a homogeneous background and a highly textured foreground object receive identical token budgets. Because the complexity of the trans-

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former’s self-attention mechanism scales quadratically with the sequence length [9], this uniform dense tokenization forces the network to waste significant computational resources processing redundant spatial information.

To address this, recent approaches have revisited classical hierarchical structures. In particular, quadtree-based spatial partitioning in ViT architectures, such as the Adaptive Patch Transformer (APT) [3], has shown promise. These models use an entropy-driven quadtree to adjust patch sizes according to local complexity, allowing spatially redundant regions to be represented more compactly. However, this line of work has so far been limited to discriminative tasks such as image classification or regression [7,3].

The application of quadtree-induced dynamic sequences to generative or reconstructive tasks, such as autoencoding, remains largely unexplored due to a critical architectural bottleneck. Generative models must ultimately map latent representations back to a dense, fixed-resolution pixel space. A quadtree encoder produces a dynamic, variable-length sequence of patch tokens that lacks a uniform spatial grid structure. Conventional ViT decoders are incapable of directly consuming these dynamic inputs, as they rely on fixed sequence lengths and static positional embeddings to reconstruct spatial relationships.

To bridge this gap, a compute-efficient autoencoder framework is introduced that pairs an APT-style quadtree encoder with a standard ViT decoder. The quadtree structure partitions the input image based on local information density, extracting fine-grained patches for high-entropy regions and coarse patches for uniform areas. To resolve the dynamic-sequence bottleneck during reconstruction, a repatchification mechanism based on learned mask tokens is introduced. The decoder receives the compressed token sequence and inserts these placeholders into the spatial locations bypassed by the encoder, restoring the sequence to a uniform grid. This design targets encoder-side efficiency while preserving a dense decoder interface. Quantitative evaluations show that the computational footprint is reduced while useful reconstruction quality is maintained.

More specifically, the contributions of this paper are threefold:

1. A quadtree-tokenized ViT autoencoder is formulated to replace uniform patchification with entropy-driven adaptive token allocation for reconstruction.
2. A repatchification operator is introduced to lift a variable-length quadtree token sequence back to a dense decoder lattice using learned mask tokens, enabling a standard ViT decoder to reconstruct adaptive-token inputs without architectural redesign.
3. An efficiency-oriented empirical study is provided, including ablations on quadtree routing and perceptual supervision, to clarify the compute–quality trade-off of adaptive tokenization for autoencoding.

2. Related Work

2.1. Adaptive Spatial Partitioning in Vision Transformers

To mitigate the quadratic cost of self-attention [9] in standard Vision Transformers [4], several methods have sought to dynamically reduce the sequence length. Early token sparsification strategies, such as DynamicViT [7], identify and entirely discard uninformative tokens during the forward pass. While effective for sequence reduction, this hard dropping mechanism irreversibly destroys spatial context, creating representational blind

spots that severely hinder dense prediction or reconstructive tasks. To overcome this limitation, more advanced approaches utilize quadtrees to merge patches dynamically rather than discarding them. The Adaptive Patch Transformer (APT) [3] assigns variable patch sizes based on local spatial complexity. To support this dynamic structure, APT projects different patch sizes into a unified embedding dimension using a zero-initialized multi-layer perceptron (ZeroMLP) and transitions from traditional batch-wise execution to a patch-wise training paradigm for efficient processing of variable-length sequences. The present framework adopts these ideas to retain an efficient ViT encoder while extending the architecture to image reconstruction.

2.2. Efficient Autoencoding and Mask Token Reconstruction

While APT provides an effective content-adaptive encoding strategy, adapting its variable-length output for generative autoencoding introduces a fundamental structural mismatch. Standard autoencoding architectures typically require the encoder and decoder to operate on a fixed, uniform token sequence. As Vector Quantization (VQ) gained prominence for reconstruction and generation tasks, several strong models emerged to improve latent representation quality and codebook usage. FVQ [1] addresses codebook collapse through a scalable training method designed to guarantee 100% codebook utilization. Soft VQ-VAE [2] instead uses a continuous relaxation that smooths the optimization landscape. However, these VQ-based methods remain dependent on fixed token grids produced by their backbones. Consequently, while they improve the quantization bottleneck, they do not directly address the spatial redundancy present in self-attention layers.

To resolve the structural mismatch between an adaptive encoder and a dense pixel space, the repatchification strategy draws conceptual inspiration from Masked Autoencoders (MAE) [6]. MAEs demonstrated that ViT decoders can reconstruct full images from a sparse set of encoded patches by filling the missing sequence positions with learned, shared mask tokens. While MAEs randomly mask patches as a self-supervised pre-training objective, the present framework deterministically bypasses low-entropy patches to prioritize computational efficiency during standard autoencoder inference. The goal is not to introduce a sparse decoder, but to provide a simple interface that lets a standard dense decoder operate on adaptive encoder outputs.

3. Methodology

The overall architecture of the proposed Quadtree-ViT Autoencoder (Q-ViT) is illustrated in Figure 1. The key distinction of Q-ViT from prior adaptive-token ViTs is not only the entropy-driven quadtree encoder, but the explicit repatchification operator that converts the resulting variable-length latent sequence into a dense token lattice that can be consumed by a conventional ViT decoder. The end-to-end pipeline therefore consists of four stages: (1) entropy-driven quadtree patchification, (2) adaptive token embedding and encoding, (3) repatchification into a dense grid via learned mask tokens, and (4) dense reconstruction with a standard ViT decoder.

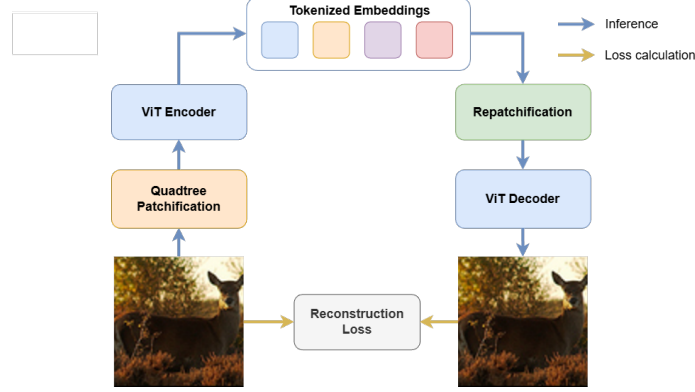


Figure 1. Overall architecture of the proposed Quadtree-ViT Autoencoder (Q-ViT), detailing the end-to-end pipeline from quadtree patchification to the calculation of reconstruction loss.

3.1. APT-Style Quadtree Patchification

As illustrated in the detailed pipeline of Figure 2, a top-down multi-scale selection strategy is applied to each input image. Each spatial region is assigned to the coarsest available patch scale at which the local Shannon entropy $H(R)$ falls below a predefined threshold τ_H :

$$H(R) = -\sum_b p_R(b) \log_2 p_R(b) < \tau_H \quad (1)$$

This produces patches of variable spatial extents depending on local complexity: uniform areas yield large, coarse patches, while detail-rich regions remain as fine-grained base patches. During this extraction step, the original spatial dimensions and grid coordinates of each patch are explicitly saved.

To project these variable-sized patches into a common embedding dimension d , the embedding strategy proposed in APT is employed. The smallest base patches are directly projected using a standard patch embedding layer. However, for larger merged patches, a dual-path embedding strategy is utilized to preserve both macro and micro structural details. A larger patch R_i is processed through two simultaneous branches. It is first spatially resized to the base dimension and passed through the standard embedding layer to capture holistic context. Simultaneously, the patch is split into its constituent sub-patches, which are individually embedded, spatially aggregated via a convolution layer, and processed by a zero-initialized multi-layer perceptron (ZeroMLP). The outputs of these two branches are combined via element-wise addition to form the final embedding t_i :

$$t_i = \text{Embed}(\text{Resize}(R_i)) + \text{ZeroMLP}(\text{Conv}(\text{Embed}(\text{Split}(R_i)))) \quad (2)$$

This content-adaptive merging and embedding process reduces the total number of patches from the standard dense grid count N to variable-length sequences of n tokens ($n \ll N$), without sacrificing the representational capacity of the spatially compressed

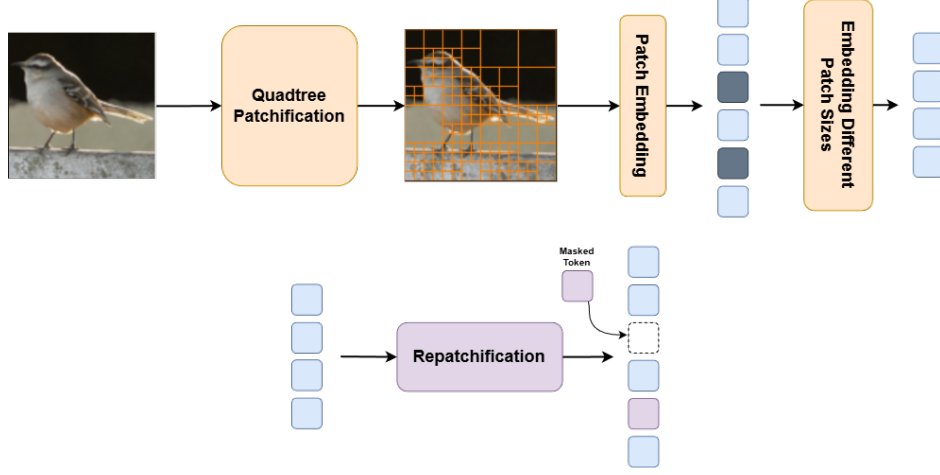


Figure 2. The internal mechanics of the patchification and repatchification modules. The encoder dynamically patchifies the image based on local entropy, while the repatchification module inserts learned mask tokens to restore the sequence to a uniform grid for decoding.

regions. Since the encoder-side self-attention cost scales quadratically with the token count, the encoder complexity is reduced from $\mathcal{O}(N^2)$ in the dense baseline to $\mathcal{O}(n^2)$ in the adaptive encoder, while the decoder remains defined on the dense lattice for faithful reconstruction.

3.2. Positional Embedding

Because the patchified tokens vary in spatial extent, a single fixed-grid positional embedding cannot be applied uniformly across scales. Instead, the encoder maintains a single set of learned absolute positional embeddings $\mathbf{P} \in \mathbb{R}^{N \times d}$ defined on the base-scale grid. For base-scale tokens, their corresponding entry in $\mathbf{P}$ is used directly. For coarser-scale tokens, $\mathbf{P}$ is spatially downsampled via bilinear interpolation to match the coarser resolution:

$$\mathbf{e}_i^{\text{pos}} = \text{Resample}(\mathbf{P}, s_i) \quad (3)$$

where s_i denotes the scale index of token i . Additionally, a learned scale embedding $\mathbf{e}^{\text{scale}} \in \mathbb{R}^{S \times d}$ is added to each token to explicitly distinguish tokens originating from different spatial scales:

$$\tilde{t}_i = t_i + \mathbf{e}_i^{\text{pos}} + \mathbf{e}_{s_i}^{\text{scale}} \quad (4)$$

This allows the encoder’s self-attention mechanism to natively interpret both the spatial location and the resolution of each token. The sequence is then processed by the standard ViT encoder blocks to produce latent representations $E = \{e_1, e_2, \dots, e_n\}$.

3.3. Repatchification and Decoding

The ViT encoder outputs a compressed sequence of n latent tokens. However, the conventional ViT decoder requires a full, uniform-grid sequence of N tokens to map the representations back to pixel space.

To bridge this gap, a repatchification step (Figure 2, bottom) restores the sequence to its maximum spatial resolution. A shared, learnable mask token $\mathbf{m} \in \mathbb{R}^d$ is defined. The $N - n$ missing patches correspond to the spatial areas that were grouped into coarser tokens during the quadtree partitioning.

For every coarse patch (where the quadtree depth is > 0), the original encoded token is mapped to its anchor location and the remaining constituent fine-grained grid positions are filled by inserting the learned mask token $\mathbf{m}$:

$$F[i, j] = \begin{cases} e_k & \text{for original encoded token locations} \\ \mathbf{m} & \text{for missing fine-grained patches} \end{cases} \quad (5)$$

Referring to Eq. 5 as the repatchification operator, it preserves the encoder’s adaptive computation while restoring the fixed spatial indexing required by the decoder. This expansion generates a complete N -sized token embedding sequence feature map F . A set of learned decoder positional embeddings $\mathbf{P}_{\text{dec}} \in \mathbb{R}^{N \times d}$ is added to every slot in F , enabling the ViT decoder to process the sequence and reconstruct the image. In this view, the decoder performs conditional dense reconstruction: retained adaptive tokens provide spatial evidence, mask tokens mark positions without direct encoded content, and self-attention propagates information across the restored lattice to infer the missing regions.

3.4. Experiment Setup

3.4.1. Dataset

The primary benchmark in this paper is STL-10, whose labeled split contains 5,000 training images and 8,000 test images. All images are resized to 224×224 so that every method operates at the same spatial resolution and can be compared directly against the ViT-B baseline under a matched decoder resolution.

3.4.2. Implementation Details

All experiments are conducted using the standard ViT-B architecture for both the encoder and the decoder across all evaluated models. Training is executed in parallel across 4 NVIDIA V100 GPUs. Unless otherwise stated, all baselines and proposed variants are trained with the same optimizer family, training schedule, image resolution, and decoder capacity so that the main difference is the tokenization strategy rather than model scale.

The models are optimized using the AdamW optimizer with a base learning rate of 1×10^{-4} and a weight decay of 0.05, utilizing a cosine learning rate schedule with a 5-epoch linear warmup over 200 training epochs. The global batch size is set to 64 across the 4 GPUs. For the proposed model, the quadtree partitioning uses 2 spatial scales with an entropy threshold of $\tau_H = 6.0$ bits. The threshold is selected on STL-10 validation performance and then held fixed within each reported experiment. This paper studies a deterministic, hand-crafted routing criterion; learning the routing policy itself is outside the scope of the present work and is discussed as future work.

3.4.3. Evaluation Scope and Limitations

The main quantitative tables focus on STL-10 because it provides a controlled benchmark for measuring the compute–quality trade-off under matched training conditions. To assess whether the efficiency mechanism is tied only to STL-10 statistics, we additionally performed auxiliary zero-shot transfer tests on CIFAR-10 using STL-10-trained checkpoints. These results should be interpreted with care. Because CIFAR-10 images are heavily upsampled to the common evaluation resolution, their absolute reconstruction metrics are not directly comparable to STL-10. CIFAR-10 is therefore used only as an auxiliary within-protocol transfer setting, summarized in Table 2. Under this setting, the proposed Q-ViT-A remained functional but trailed the dense baseline in absolute reconstruction quality. This suggests that the encoder-side efficiency mechanism transfers more readily than the current decoder-side reconstruction strategy. The STL-10 study is therefore presented as the primary controlled benchmark, while broader cross-dataset validation remains an important extension.

4. Results

The proposed compute-efficient autoencoder framework is evaluated against standard ViT-B and established discrete tokenization baselines, including Soft VQ-VAE [2] and FVQ [1]. The evaluation focuses on reconstruction quality and computational overhead (GFLOPS). STL-10 serves as the primary benchmark, and CIFAR-10 is used only for auxiliary zero-shot transfer analysis.

4.1. Quantitative Reconstruction Quality and Efficiency

As shown in Table 1, the proposed method achieves the lowest computational cost while maintaining competitive pixel-level reconstruction. By reducing redundant spatial tokens before encoding and then restoring a dense decoder lattice, the method operates at 36.21 GFLOPS, representing a 17.5% reduction in compute compared to the standard ViT-B baseline (43.88 GFLOPS).

The proposed approach improves on FVQ and remains competitive with Soft VQ-VAE in pixel-level metrics, reaching a PSNR of 29.60 and an SSIM of 0.8502 without the parameter cost of discrete codebooks. However, all three recent efficient autoencoding methods in Table 1 show lower PSNR and SSIM than the dense ViT-B baseline. This suggests that the drop in pixel-level reconstruction quality is not unique to the proposed method, but reflects a broader limitation of current efficiency-oriented autoencoder designs. A trade-off in perceptual quality is also observed, with an LPIPS score of 0.1837. This likely occurs because the learned mask tokens must help reconstruct regions that were not encoded at the finest scale.

4.2. Qualitative Results

Figure 3 provides a visual comparison of reconstruction quality across the evaluated methods. Column (a) illustrates the entropy-based quadtree patchification. The algorithm allocates fine-grained patches to high-frequency foreground objects (e.g., the rider, the

Table 1. Reconstruction quality and computational cost comparison across methods.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	GFLOPS $\downarrow$
ViT-B [5]	34.03	0.9340	0.1338	43.88
FVQ [1]	24.72	0.7017	0.2799	406.4
Soft VQ-VAE [2]	26.96	0.7694	0.1663	117.4
Proposed Method	<u>29.60</u>	<u>0.8502</u>	0.1837	36.21

deer, and the cat’s face) while grouping homogeneous background regions (e.g., the sky and defocused walls) into large, coarse patches.

As seen in Column (e), the proposed method preserves sharp textures and edges in dense foreground regions better than the FVQ (c) baseline and competitively with Soft VQ-VAE (d). At the same time, the qualitative results also show the source of the higher LPIPS score in the quantitative analysis. In homogeneous background regions where the encoder drops spatial information, the decoder’s reliance on a shared learned mask token can produce visible block-like artifacts. This illustrates the main trade-off of the architecture: lower computation at the cost of weaker perceptual smoothness in skipped regions.

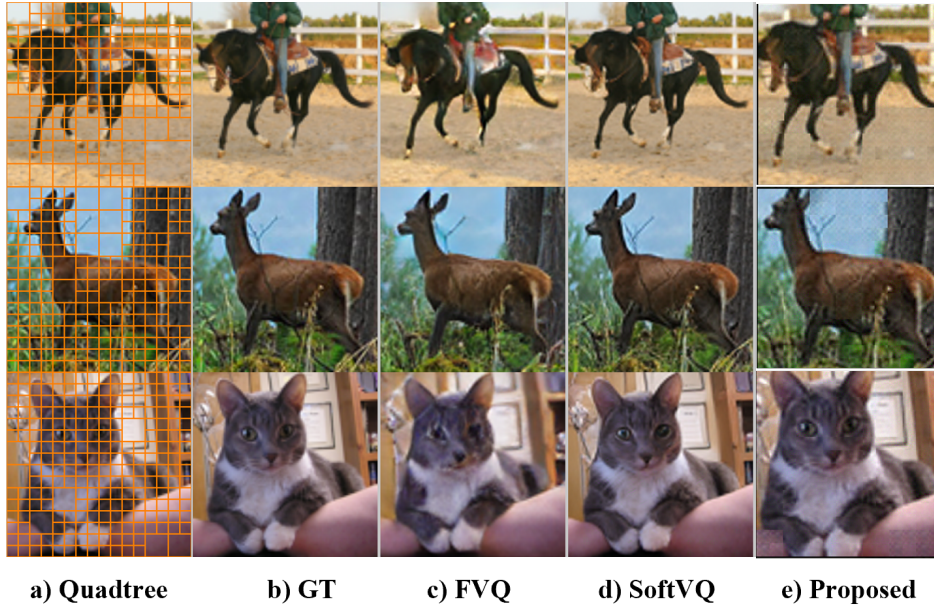


Figure 3. Qualitative reconstruction results. (a) Proposed quadtree patchification overlay, (b) Original image, (c) FVQ, (d) Soft VQ-VAE, and (e) Proposed Method. The proposed method preserves high-frequency details in the foreground while introducing blocky artifacts in the masked background regions.

4.3. Parameter Count

The adaptive patchification and repatchification mechanisms introduce negligible parameter overhead to the standard ViT architecture. The only additional learnable parameters are the multi-layer perceptron (MLP) used for mixed-scale patch embedding and

the singular shared mask token $\mathbf{m}$ utilized during decoding. Thus, the proposed method achieves significant efficiency improvements without a meaningful increase in model capacity, indicating that the observed gains arise from computation reallocation rather than from a larger model.

4.4. Auxiliary Zero-Shot CIFAR-10 Transfer

Table 2 reports auxiliary zero-shot transfer results on CIFAR-10 using STL-10-trained checkpoints. These values are included to characterize transfer behavior within a common CIFAR-10 protocol, not to compare absolute performance directly against STL-10. Within this setting, the dense baseline remains strongest overall. Relative to FVQ, the proposed Q-ViT-A remains competitive in PSNR and improves SSIM, but still lags in LPIPS. This further supports the observation that recent non-dense reconstruction methods can remain useful while still trailing the dense baseline in some quality metrics.

Table 2. Auxiliary zero-shot CIFAR-10 transfer results using STL-10-trained checkpoints. Because CIFAR-10 images are strongly upsampled under the common evaluation protocol, these numbers are intended for within-CIFAR comparison only and not for direct comparison against STL-10 results.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ViT-B baseline	43.63	0.9804	0.1486
FVQ	30.61	0.9352	<u>0.1773</u>
Proposed Q-ViT-A	<u>36.29</u>	<u>0.9417</u>	0.3210

5. Ablation Study

To isolate the contributions of the proposed structural modifications and training objectives, an ablation study is conducted on the quadtree partitioning and the integration of a perceptual loss function.

Without Quadtree Patchification: Removing the entropy-driven spatial partitioning forces the model to encode all patches at the finest resolution, strictly mirroring a standard ViT-B autoencoder. This configuration maximizes the token count, thereby eliminating the inference speed advantages and GFLOPS reduction, establishing the upper bound for reconstruction quality (34.03 PSNR).

Addition of Perceptual Loss: The baseline proposed model optimizes solely for mean squared error (MSE), which can result in overly smooth reconstructions in the regions populated by the learned mask tokens, driving up the LPIPS metric. To mitigate this, an explicit LPIPS perceptual loss term [11] is introduced during training. As shown in Table 3, integrating the perceptual loss improves the LPIPS score from 0.3051 to 0.1837, generating sharper structural details in the reconstructed output. Notably, this addition also yields a significant improvement in pixel-wise metrics, increasing PSNR from 27.77 to 29.60 and SSIM from 0.7893 to 0.8500. This indicates that the perceptual gradients provide structural guidance for the learned mask tokens during repatchification.

The auxiliary zero-shot CIFAR-10 results in Table 2 also suggest that this benefit may not carry across datasets as reliably as the encoder-side efficiency gain. In the present architecture, LPIPS is applied after adaptive token reduction and repatchification, so some spatial locations are reconstructed from retained context and shared mask

tokens rather than from direct fine-scale evidence. Under distribution shift, this can make perceptual supervision less stable: LPIPS may favor plausible feature-space completions even when they are not the most faithful pixel-level reconstructions. This interpretation is consistent with the CIFAR-10 transfer results, where the Q-ViT_A remained competitive in PSNR and improved SSIM over FVQ, but still lagged in LPIPS. LPIPS in Q-ViT_A is therefore best viewed as a conditional refinement term whose effect depends on the routing regime and dataset statistics, rather than as a uniformly beneficial objective.

Table 3. Ablation of proposed quadtree components and loss functions.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	GFLOPS $\downarrow$
ViT-B (No Quadtree)	34.03	0.9340	0.1338	43.88
Proposed (Quadtree + MSE)	27.77	0.7893	0.3051	36.21
Proposed (Quadtree + MSE + LPIPS)	<u>29.60</u>	<u>0.8500</u>	<u>0.1837</u>	36.21

6. Conclusion

This paper introduces a compute-efficient autoencoder framework that pairs an Adaptive Patch Tokenization (APT) encoder with a standard ViT decoder. By dynamically scaling sequence lengths according to local image entropy, the architecture bypasses redundant processing in homogeneous regions before the encoder. The core methodological contribution is the repatchification operator, which bridges the variable-length encoder output back to a uniform spatial grid using learned mask tokens, allowing a conventional ViT decoder to reconstruct adaptive-token inputs without architectural redesign.

The resulting model achieves a 17.5% reduction in GFLOPS compared to standard ViT autoencoders while maintaining competitive pixel-level reconstruction on STL-10. The present study also highlights four practical limitations: the reliance on a fixed entropy threshold, the fact that decoder computation remains dense after repatchification, the current concentration of full benchmark results on a single primary dataset, and the sensitivity of perceptual supervision under transfer. In particular, the PSNR and SSIM gap against the dense ViT-B baseline remains an important limitation. Since recent VQ-based baselines also fall below the dense ViT-B baseline in these metrics, this gap appears to be a broader challenge for efficient autoencoder designs rather than an issue isolated to Q-ViT_A. As shown by the auxiliary zero-shot CIFAR-10 results in Table 2, the proposed Q-ViT_A remained functional under transfer and improved over FVQ in PSNR and SSIM, but still lagged in LPIPS and did not match the dense baseline. These observations suggest that encoder-side efficiency transfers more reliably than perceptual reconstruction quality, and that the current repatchification-based decoder remains a bottleneck under distribution shift. Future work will therefore focus on learnable routing mechanisms, improved decoding of missing regions, better-conditioned perceptual supervision, more extensive multi-dataset validation, and extensions of the variable-size decoding strategy to 3D quadtrees for video processing.

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